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What is an Ethical Engineer?

Engineering is a weird concept. Many people are divided into what they believe. Some people only take ideas into account that we know are real (The idea of doing something bad can affect someone's life, a car can go fast because of the engine, etc.). There are others that believe in the world of science where things are very theoretical (Electric Charge distributions on an infinite plane, Ideal components to a product, etc.). People who go into a non-stem degree but deal with products of stem, often get the general idea of how the product works, and can hold a well thought conversation about it on their own. A scientist will be able to go up and beyond talking about the theory behind it, they are not able to create it but can still hold their own on this matter. The Engineer is the middle person in all this; they have to know the theory and the reality of the situation. This is a very important concept in the advanced technological world that we live in today. Nowadays, people feed off of the eager to have new, more efficient technologies that Engineers have to design in a relatively fast manner. Due to this demand, not only by the consumer, but by the corporations that fund these designs, Engineers have to make a lot of decisions that deal with ethical reasoning. In this paper, the discussion will be about different Ethical backgrounds that Engineers have to face on a daily basis, and towards the end of the discussion, we will discuss what I think an Engineer should really resemble to make them an Engineer. We will start off with Ethical theories, Deontology, Consequentialism, and Virtues.

Deontology

Deontology, by definition, is a theory that offers rules to differentiate right from wrong. There are many different types of Deontology. Two key topics that were a huge focus in the course were Agent Centered and Patient Centered Deontology. Agent Centered is all about how oneself upholds their moral responsibility when doing tasks. Whereas, Patient centered, is all about the affect your work may have on people's rights. These two topics are very good complements to each other. As humans, we have a lot of different thoughts every day. The human brain is a very interesting concept that is very difficult to understand. Engineers have to use their brain all the time to solve the world's technological problems. However, this is something that can be difficult to do as an Engineer. We as Engineers will have projects, depending on the company, that we may disagree with because our Agent centered ethics says it does not fit our moral rule set, we may even have the other opinion of a Patient Centered belief that what we are doing as Engineers can have a bad outcome on people's lives. As an Example, we can talk about Engineering style situations and how an Engineer may go about the problem. Let's say I am an Engineer working for a big company. I have my dream job, with aspirations to climb the corporate

ladder, and become a wealthy individual. However, I am still at the bottom of the totem pole working with a design team. Our boss has given us a new project to work on, that if done correctly, could help me reach my goals, but at the same time this product could affect people in a very bad way. With an Agent Centered focus, this would be a good scenario for an Engineer to be in. They are being challenged with a new product from their boss, which can lead them to success. The question that we must think about is does this align with the Engineers morals. Well, that is a tricky thing to unpack. Do I (the Engineer) think that that this product would do good and help myself reach the path of my goals? Do I deem the work that I am doing morally right? These are questions that an Agent Centered could ask. Notice I did not say anything about anyone else involved. These are self-dilemmas that an Engineer has to think about. An opinion on this is that this is a very selfish way of thinking, however, as humans, we all have thoughts like this of will it benefit me. I will go more into how this topic may seem good soon. I would like to bring up the Patient Centered side. For Patient Centered, things that I may think about are, how does this product affect the people that will consume it? How does this product affect the company I work for? Notice how the Engineer (me) is not asking questions of how it could affect their life but rather giving the attention to how this product will affect others. This gives the challenge of the fact that your organization is asking you to complete this task for the benefit of the company, however the people getting affected are also the consumer, so the Engineer needs to take this into account. Both of these centered approaches are difficult to do sometimes given the situation.

This leads me to bring up two other forms of Deontology, Kant and Contractarian Ethics. Kant's belief is that humans should uphold universal moral laws such as not lying, stealing, cheating, etc. Whereas Contract ethics states we as humans are not perfect and that we make mistakes. Contracts are also about social agreements like Government rules and religious beliefs. I bring these up to show that Deontology is very complex. Just because someone is more Agent Centered does not mean that they are selfish individuals as an Engineer, it just means that they care about their actions and achieving their moral goals. On the Contrary, Patient ethics sounds better because it sounds more selfless, but it doesn't have to be. The point I am trying to make here is If Engineers make certain social contracts, follow the rules of the work place, mix in some of their own personal morals, and think about the others that get affected, Deontology wouldn't be that hard for a Engineer to bring about into their environment.

Consequentialism

Consequentialism by definition is, “Of all the things a person might do at any given moment, the morally right action is the one with the best overall consequences”. This definition seems a little bland, so let’s unpack it. Consequentialism is not really about trying to do strictly the right thing, it is more about how I can do the right thing morally, with minimal pushback from people who oppose my mission. Engineers have to face this a lot in the workforce. A great example is designing a new product that almost everyone uses nowadays, such as a car or a gaming console. Engineers are the ones who design the product not what it will look like necessarily, but how it functions. If we do not make a good product no one is going to buy it, and the company will go under. A good example of this is making safety systems for cars. Engineers may design the sensors that let the driver know they are moving too far left or right into another lane, and that signal may be sent to another system that helps coast the car back into the lane it should be in. These systems are rather new in cars and are quite enjoyed by many users. However, I am a part of the opposition. I did enjoy it at once, but when I was driving the car, the sensor thought I was veering off the road I was on and actually forced my car to move a bit into a jersey barrier on the highway. Luckily the passengers and I were okay. Now, this is something I and I am sure many others can complain about since it is a fault in the sensor system from this car brand that I am leaving anonymously. This is something an Engineer would have to take into account when designing in the future. However, if an Engineer looks at the statistics and believes in Consequentialism, this minor sensor issue may not look all that bad, because if we are looking at stats, this only happens every so often, and therefore the engineer is still making a net positive. This is something that can be looked at as good or bad with Consequentialism. Good in the sense that not many cars have issues with this for that make and model, and bad in the sense that there are still problems with some of these sensors and can risk someone’s life. This would fall under the category of Plain Scalar Consequentialism. There are many other forms such as Reasonable, Expected, or Dual Consequentialism. They are all pretty similar with a few differences. Reasonable and expected, are more about how the consequences of our actions need to be reasonably good overall. Dual Consequentialism states that there really isn’t a definition for what is good or bad, and it really is just subjective. I think all of these have their place in Engineering, but I think the overarching idea as a whole, is that Engineer are going to have to make a product, some people are going to like it, some people are going to hate it, and some people are going to feel indifferent about it. What the engineer will want is that the net outcome is positive, and thus the Engineer will feel good about their design. This is still bad in some respects. If you take a more negative approach in life, then people will realize that Consequentialism can be abuse by lazy engineers would are just there to collect their

paycheck. However, that doesn't mean there is no light at the end of the tunnel. There are instances that people actually use consequentialism the proper way, by actively trying to make a net good, and just have to deal with minor pushbacks, that they can try and use to make their design better for the next version.

Virtues

Virtue Ethics by definition, a desirable characteristic or trait of an individual. This theory in my opinion is more of personal ethical idea, which can show the true colors of people. Virtue ethics is a form of ethics that is formed for oneself by many factors in everyday life. The factors can be cultural, family based, experience based, etc. The point I am getting at is for virtue ethics, there is no set in stone way of describing virtues, it something that is molded into you is how someone can make their day-to-day decisions. This is something worth noting in Engineering. Engineering is very diverse in what it can do and has a lot of different people with different backgrounds in it. These backgrounds can make a difference in how something was done because at the end of the day our brain will subconsciously make our decisions based off of these virtues that we have. As Engineers, we are going to have to do a lot of work, and in order to do that work, a lot of engineers will need certain virtues, like diligence, loyalty, and honesty. Diligence will give the Engineer the ability to stay focused and be able to complete all tasks that are assigned to them. This is crucial when working on big projects that the consumer wants and the company has a short deadline for. Loyalty is good not only for the company but to the people the Engineering firm are helping. Loyalty shows you stay true to the firm's values, want to complete the work that they give you, as well as it shows that you care about the consumer, and will like to give them the best product possible. Honesty is all about staying true to others and also to your own moral compass. This is very good for an Engineer to have this trait because it shows that they care about the project. Engineers will maybe earn these style values maybe from parents, or someone they look up to. It does not necessarily mean though that it will come from that. One of the forms of virtue ethics we learned about was Confucian Ethics. Confucius believed that virtue ethics was important in the sense that people shouldn't just follow the rules, but they should have good character traits. This can bring up a tough answer, of what good means. That is a very tough and opinionated question again stemming from our virtues and backgrounds. Overall, there have been societal beliefs about what good virtues are like, honesty, courage, respect, etc. This style of ethics proves why this is important for engineers. Confucius states that we cannot just follow a rule book, that we still need to be good human beings. Engineers, although they can follow the rules (deontology), have a net positive (Consequentialism), they must still

have virtues holding them back from bad decisions or virtues that push them to make the right decision. If done properly Virtues can be a very good asset to an Engineer.

Conclusion

Engineers are the pinnacle to the ever-changing world. Due to this, we have a lot of responsibilities. This can mean a collection of non-ethical and ethical things. In this paper we discussed all the different Ethics we have learned this far. My final response is what I believe an ethical Engineer is. An Ethical Engineer in my opinion does not simply use only one form of ethics to make their decisions. I believe in a universe where all of the ethics are correct, but they have their place given the situation. There will be times where an Engineer will not know if their future creation will meet certain specifications, so they will read the rule book (deontology) to make sure they are making the best decisions. There will be times when an Engineer will have to make tough decisions for a project, when the project is created, the engineer will have pushback and will have people that are for it. They must try to make the best outcome decision possible in their respective field and projects. For Virtue Ethics. The Engineer will be faced with problems in their job that will require them to make a decision that was ultimately decided by their virtues, on what they believe is the correct way to go about this, regardless of the outcomes, and rule books. To be an Ethical Engineer means to me that the Engineer given any problem will do everything in their power to uphold good morals, follow the rules, and make the best outcome for the world. Although this sounds vague, I think this is something a lot of people can agree with and should be a respected way of thought for the Engineering world.

Citations

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